

VI Semester B.E. Examination
(Common to E&C/E&C-II)
Automotive Electronics (24EECC305/24EEIC316)

Duration: 3 hours

Max. Marks: 100

Note: i) Answer any TWO full questions from UNIT-I, any TWO full questions from UNIT-II and any ONE full question from UNIT-III.

UNIT-I

- 1
 - a. Explain the V-design methodology applied to embedded software development for Electronic Control Unit. (08marks)
 - b. The engine abruptly encounters a rapid spike in cylinder pressure followed by sharp oscillations in cylinder pressure. How this situation is handled by an engine management system and provide the corresponding closed loop control strategy. (06marks)
 - c. Determine the fuel injector pulse duration and fuel quantity for the four cylinder fully warmed up and very cold engine running at 2000 rpm with a fuel flow rate of 0.014 kg/sec and mass air flow rate of 0.02 kg/sec. (06marks)
- 2
 - a. Develop a fuel control strategy for varying operating modes for demonstrating their effects on efficiency, emission and drivability in conditions like coldstart and idle. (08marks)
 - b. Explain the construction and working principle of MAP sensor. (06marks)
 - c. Explain the role of market drivers for automotive industry. (06marks)
- 3
 - a. Explain the need of secondary injection system in the engine management system. (08marks)
 - b. Illustrate the engine control system for NO_x emission control. (06marks)
 - c. Explain the construction and working principles of fuel injector actuator. (06marks)

UNIT-II

- 4
 - a. The fast moving car is turning at the corner, if the vehicle is turning less/more than the driver's intention. Suggest and explain a suitable control system along with brake circuit configuration to overcome this condition. (08marks)
 - b. Calculate nominal and maximum T_{HEADER} , $T_{RESPONSE}$ and T_{FRAME} , if LIN protocol is operating at 20 kbps baud rate and reserved time is set to 30% for transmitting eight bytes of data. (06marks)
 - c. Illustrate the role of electronic suspension system for ride and handling by controlling the damping co-efficient. (06marks)
- 5
 - a. Explain the protocol frame format suitable for video signal with low latency and high fidelity for parking assistance system. (08marks)
 - b. Explain the communication architecture of FlexRay protocol. (06marks)
 - c. Discuss the physical mechanism of wheel lock and vehicle skid, that can occur during panic braking. Also calculate wheel slip for front left and front right wheel for which the corresponding speed is measured as 60 kmph and 62 kmph, when the vehicle speed is 68 kmph. (06marks)
- 6
 - a. Assume that the driver has activated the cruise control system for the desired speed of 80 kmph. Explain the action of throttle and its control for the following conditions, (08marks)
 - i) If the car traveling on a level road.
 - ii) If the car is traveling on uphill road.

b. Explain the differences between classical CAN and CAN FD. (06marks)

c. Illustrate the frame format of automotive ethernet protocol. (06marks)

UNIT-III

7 a. Determine the ASIL value for an ABS system failure as per the ISO 26262 standards. (08marks)

b. Explain the ISO26262 functional safety principles. (06marks)

c. Explain the SAE levels of automation. (06marks)

8 a. Illustrate the methodology, that allows extraction of diagnostic trouble codes and real time data. (08marks)

b. Explain the KWP2000 protocol. (06marks)

c. Explain the on-board diagnostic stack with an example. (06marks)